

Erdős Problem #1131

EP-1131

L1 - TRACTABLE

PARTIALLY SOLVED

Combinatorics

ORIGINAL PROBLEM STATEMENT

For $x_1, \dots, x_n \in [-1, 1]$ let $l_k(x) = \frac{\prod_{i \neq k} (x - x_i)}{\prod_{i \neq k} (x_k - x_i)}$, which are such that $l_k(x_k) = 1$ and $l_k(x_i) = 0$ for $i \neq k$.

What is the minimal value of $I(x_1, \dots, x_n) = \int_{-1}^1 \sum_k |l_k(x)|^2 dx$? In particular, is it true that $\min I = 2 - (1 + o(1)) \frac{1}{n}$?

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>